

MAT 527: Functions of a Complex Variable I

Submission II

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May 24, 2026

1. Theorem If $\mathcal{F}_1$ and $\mathcal{F}_2$ are closed, then $\mathcal{F}_1 \cup \mathcal{F}_2$ is closed.

Proof One: The complement of $\mathcal{F}_1 \cup \mathcal{F}_2$ is $\mathbb{C} - \mathcal{F}_1 \cup \mathcal{F}_2 = (\mathbb{C} - \mathcal{F}_1) \cap (\mathbb{C} - \mathcal{F}_2)$, by DeMorgan's Law. Since $\mathcal{F}_1$ and $\mathcal{F}_2$ are closed, the sets $\mathbb{C} - \mathcal{F}_1$ and $\mathbb{C} - \mathcal{F}_2$ must be open. The intersection of two open sets is also open; therefore, $\mathbb{C} - \mathcal{F}_1 \cup \mathcal{F}_2$ is open. Thus, $\mathcal{F}_1 \cup \mathcal{F}_2$ is closed. ■

Proof Two: Let (z_n) be a convergent sequence with terms in $\mathcal{F}_1 \cup \mathcal{F}_2$ whose limit is $L \in \mathbb{C}$. By the Pigeon-hole Principle, at least one of $\mathcal{F}_1$ and $\mathcal{F}_2$ contains infinitely many terms of (z_n) . Without loss of generality, suppose $\mathcal{F}_1$ contains infinitely many terms of (z_n) . These terms form a subsequence (z_{n_k}) of (z_n) . But (z_{n_k}) converges to L , since subsequences of convergent sequences converge to the limit of the parent sequence. Since $\mathcal{F}_1$ is closed, L must be an element of $\mathcal{F}_1$. Therefore, $L \in \mathcal{F}_1 \cup \mathcal{F}_2$. Thus, $\mathcal{F}_1 \cup \mathcal{F}_2$ is closed. ■

2. Theorem If $\{\mathcal{F}_\lambda\}_{\lambda \in \Lambda}$ is a family of closed sets, then $\bigcap_{\lambda \in \Lambda} \mathcal{F}_\lambda$ is closed.

Proof One: The complement of $\bigcap_{\lambda \in \Lambda} \mathcal{F}_\lambda$ is $\mathbb{C} - \bigcap_{\lambda \in \Lambda} \mathcal{F}_\lambda = \bigcup_{\lambda \in \Lambda} (\mathbb{C} - \mathcal{F}_\lambda)$, by DeMorgan's Law. Since each $\mathcal{F}_\lambda$ is closed, the sets $\mathbb{C} - \mathcal{F}_\lambda$ are open. This makes $\mathbb{C} - \bigcap_{\lambda \in \Lambda} \mathcal{F}_\lambda$ a union of a family of open sets, and, therefore, open. Thus, $\bigcap_{\lambda \in \Lambda} \mathcal{F}_\lambda$ is closed. ■

Proof Two: Let (z_n) be a convergent sequence with terms in $\bigcap_{\lambda \in \Lambda} \mathcal{F}_\lambda$ whose limit is $L \in \mathbb{C}$. Then, for all $\lambda \in \Lambda$, and all $n \in \mathbb{N}$, $z_n \in \mathcal{F}_\lambda$. Since each $\mathcal{F}_\lambda$ is closed, it must be the case that for all λ , $L \in \mathcal{F}_\lambda$. Therefore, $L \in \bigcap_{\lambda \in \Lambda} \mathcal{F}_\lambda$. Thus, $\bigcap_{\lambda \in \Lambda} \mathcal{F}_\lambda$ is closed. ■